

1. (10 points) Suppose that the vector $(x, 20, 26)$ is in the span of $(1, -1, 3)$ and $(2, -1, 4)$. Find x .

Solution. $x = 66$.

2. Let A be a 4×5 matrix, and let $\mathbf{b}_0 \in \mathbb{R}^4$. Suppose that $A\mathbf{x} = \mathbf{b}_0$ has standard solution

$$\mathbf{x} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 0 \\ -4 \\ 1 \end{bmatrix}.$$

(Note that there is no variable in front of the first vector.)

- (a) (3 points) Fill in the blank (no explanation needed): $\text{rank}(A) = \underline{\quad 2 \quad}$.

Solution. The number of free variables is 3, so the number of pivot variables is $5 - 3 = 2$. By definition, the rank of a matrix is the number of pivots in its (reduced) echelon form.

- (b) (2 points) Circle one (no explanation needed): Every vector $\mathbf{b} \in \mathbb{R}^4$ is a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5$, where $\mathbf{a}_j$ denotes the j -th column of A .

True

False

Solution. The statement is equivalent to saying that $A\mathbf{x} = \mathbf{b}$ is consistent for all $\mathbf{b}$. For any $m \times n$ matrix, this is true if and only if $\text{rank}(A) = m$ (i.e., the echelon form of A has no rows of zeros). In this case, $\text{rank}(A) = 2$ and $m = 3$, so the statement is false.

- (c) (5 points) Find scalars $c_1, c_2, c_3, c_4 \in \mathbb{R}$ such that $\mathbf{a}_5 = c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + c_3\mathbf{a}_3 + c_4\mathbf{a}_4$.

Solution. Each of the vectors

$$\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 0 \\ -4 \\ 1 \end{bmatrix}$$

is a solution to $A\mathbf{x} = \mathbf{0}$. This follows from the general fact that if $\mathbf{u}$ and $\mathbf{v}$ are both solutions to $A\mathbf{x} = \mathbf{b}$, then $\mathbf{u} - \mathbf{v}$ is a solution to $A\mathbf{x} = \mathbf{0}$. (See also Theorem 5.3 in the textbook.)

We only need the last one. It tells us that

$$A \begin{bmatrix} -3 \\ 0 \\ 0 \\ -4 \\ 1 \end{bmatrix} = -3\mathbf{a}_1 + 0\mathbf{a}_2 + 0\mathbf{a}_3 - 4\mathbf{a}_4 + \mathbf{a}_5 = \mathbf{0}.$$

Rearranging gives $\mathbf{a}_5 = 3\mathbf{a}_1 + 4\mathbf{a}_4$, so we can take $c_1 = 3$, $c_2 = c_3 = 0$, $c_4 = 4$. (There are other possibilities.)